EXPERIMENT 6

Kartik Bhat	Roll: No - 03
Spandan Deb	Roll: No - 13

Aim: Interfacing of Components in Sound Level Detector Using Electret Microphone.

Theory:

Unit Testing of Electret Microphone with Arduino

Connections:

Electret Microphone Pin	Arduino Pin
Signal Pin	A0
Ground Pin	GND

Code for Testing Electret Microphone

```
#define MIC_PIN A0  // Define the pin where the microphone is connected


void setup() {
  Serial.begin(9600);  // Initialize serial communication for debugging
  pinMode(MIC_PIN, INPUT);  // Set the microphone pin as input
}


void loop() {
  int sensorValue = analogRead(MIC_PIN);  // Read the value from the microphone
  Serial.println(sensorValue);  // Print the value to the Serial Monitor
  delay(100);  // Small delay for readability in the Serial Monitor
```

}

Unit Testing of LED with Arduino

Connections:

LED	Arduino
Anode	DO
Cathode	GND

Code for Testing LED

```
#define LED_PIN 13  // Define the pin where the LED is connected

void setup() {
  pinMode(LED_PIN, OUTPUT);  // Set the LED pin as an output
  Serial.begin(9600);       // Initialize serial communication for debugging
}

void loop() {
  digitalWrite(LED_PIN, HIGH);  // Turn the LED on
  Serial.println("LED is ON");
  delay(1000);              // Wait for 1 second

  digitalWrite(LED_PIN, LOW);   // Turn the LED off
  Serial.println("LED is OFF");
  delay(1000);              // Wait for 1 second
}
```

Final Integration:

Final Circuit Connections:

Signal Pin->A0 on Arduino

Ground Pin ->GND on Arduino

LED

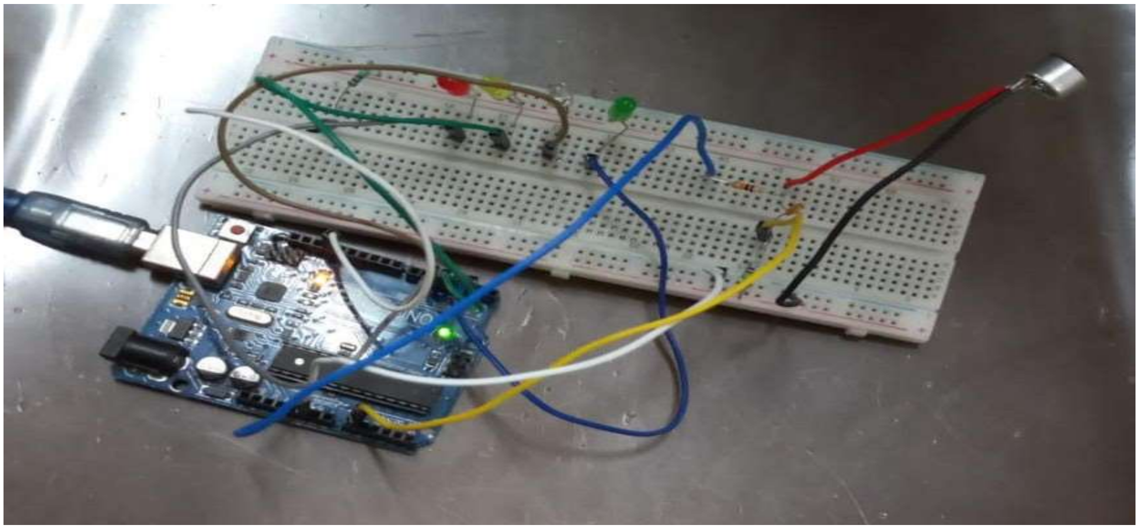
Anode ->DO on Arduino

Cathode ->GND on Arduino

Capacitor (0.1 μF) between Microphone Signal Pin and GND
Resistor (10 kΩ) between Microphone Signal Pin and 5V

Final Code:

```
#define MIC A0

int sig = 0;

void setup() {
 pinMode(2, OUTPUT);
 pinMode(3, OUTPUT);
 pinMode(4, OUTPUT);
 pinMode(5, OUTPUT);
}

void led() {
 sig = analogRead(MIC)*50;

 if (sig>1)  {digitalWrite(2, HIGH);} else {digitalWrite(2, LOW);}
 if (sig>300) {digitalWrite(3, HIGH);} else {digitalWrite(3, LOW);}
 if (sig>800) {digitalWrite(4, HIGH);} else {digitalWrite(4, LOW);}
 if (sig>950) {digitalWrite(5, HIGH);} else {digitalWrite(5, LOW);}
}

void loop() {
 led();
}
```



Working of the System

1)Sound Capture: The electret microphone captures sound waves from the environment and converts them into electrical signals.

2)Signal Transmission: These electrical signals are sent to the analog input pin (A0) on the Arduino for processing.

3)Signal Processing: The Arduino reads the analog signal using the analogRead() function and processes it to determine the sound intensity.

4)Threshold Comparison: The processed signal is compared against predefined thresholds to determine the sound level.

5)LED Control: Based on the comparison, the Arduino controls the LEDs connected to its digital output pins (e.g., pins 2, 3, 4, 5).

6)Real-Time Response: The system continuously monitors sound levels in real-time, updating the LEDs to reflect changes in sound intensity.

7)Visual Feedback: The LED indicators provide visual feedback, allowing users to observe the current sound level

Conclusion:

In this experiment we successfully performed unit testing of each element Arduino .We ensured that the components are functioning properly and are in the right place for accurate connections.